

# H1050 Hardened S46500 Stainless Steel

H1050 hardened S46500 stainless steel is S46500 stainless steel in the hardened (H) condition. It has the highest ductility compared to the other variants of S46500 stainless steel.

The graph bars on the material properties cards below compare H1050 hardened S46500 stainless steel to: wrought precipitation-hardening stainless steels (top), all iron alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

## Mechanical Properties

Brinell Hardness

450

Elastic (Young's, Tensile) Modulus

190 GPa

28 x 10<sup>6</sup> psi

Elongation at Break

14 %

Fatigue Strength

|                                  |                           |
|----------------------------------|---------------------------|
| 820 MPa                          | 120 x 10 <sup>3</sup> psi |
| Poisson's Ratio                  |                           |
| 0.28                             |                           |
| Reduction in Area                |                           |
| 50 %                             |                           |
| Rockwell C Hardness              |                           |
| 49                               |                           |
| Shear Modulus                    |                           |
| 75 GPa                           | 11 x 10 <sup>6</sup> psi  |
| Shear Strength                   |                           |
| 950 MPa                          | 140 x 10 <sup>3</sup> psi |
| Tensile Strength: Ultimate (UTS) |                           |
| 1570 MPa                         | 230 x 10 <sup>3</sup> psi |
| Tensile Strength: Yield (Proof)  |                           |
| 1430 MPa                         | 210 x 10 <sup>3</sup> psi |

# Thermal Properties

|                       |  |
|-----------------------|--|
| Latent Heat of Fusion |  |
|-----------------------|--|

280 J/g

Maximum Temperature: Corrosion

680 °C

1260 °F

Maximum Temperature: Mechanical

780 °C

1430 °F

Melting Completion (Liquidus)

1450 °C

2650 °F

Melting Onset (Solidus)

1410 °C

2570 °F

Specific Heat Capacity

470 J/kg-K

0.11 BTU/lb-°F

Thermal Expansion

11 µm/m-K

# Otherwise Unclassified Properties

Base Metal Price

15 % relative

Density

|                                     |                             |
|-------------------------------------|-----------------------------|
| 7.9 g/cm <sup>3</sup>               | 490 lb/ft <sup>3</sup>      |
| Embodied Carbon                     |                             |
| 3.6 kg CO <sub>2</sub> /kg material |                             |
| Embodied Energy                     |                             |
| 51 MJ/kg                            | 22 x 10 <sup>3</sup> BTU/lb |
| Embodied Water                      |                             |
| 120 L/kg                            | 14 gal/lb                   |

# Common Calculations

|                                          |  |
|------------------------------------------|--|
| PREN (Pitting Resistance)                |  |
| 15                                       |  |
| Resilience: Ultimate (Unit Rupture Work) |  |
| 210 MJ/m <sup>3</sup>                    |  |
| Resilience: Unit (Modulus of Resilience) |  |
| 5300 kJ/m <sup>3</sup>                   |  |
| Stiffness to Weight: Axial               |  |
| 14 points                                |  |
| Stiffness to Weight: Bending             |  |

25 points

Strength to Weight: Axial

56 points

Strength to Weight: Bending

38 points

Thermal Shock Resistance

55 points

# Alloy Composition

|                 |              |
|-----------------|--------------|
| Iron (Fe)       | 72.6 to 76.1 |
| Chromium (Cr)   | 11 to 12.5   |
| Nickel (Ni)     | 10.7 to 11.3 |
| Titanium (Ti)   | 1.5 to 1.8   |
| Molybdenum (Mo) | 0.75 to 1.3  |
| Manganese (Mn)  | 0 to 0.25    |
| Silicon (Si)    | 0 to 0.25    |
| Carbon (C)      | 0 to 0.020   |
| Phosphorus (P)  | 0 to 0.015   |
| Sulfur (S)      | 0 to 0.010   |
| Nitrogen (N)    | 0 to 0.010   |

All values are % weight. Ranges represent what is permitted under applicable standards.

# Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

## Further Reading

ASTM A564: Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

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Welding Metallurgy of Stainless Steels, Erich Folkhard et al., 2012

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ASTM A959: Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels

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Properties and Selection: Irons, Steels and High Performance Alloys, ASM Handbook vol. 1, ASM International, 1993

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Corrosion of Stainless Steels, A. John Sedriks, 1996

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ASM Specialty Handbook: Stainless Steels, J. R. Davis (editor), 1994

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Advances in Stainless Steels, Baldev Raj et al. (editors), 2010

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